

Worksheet-1

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All experimental results and visualizations are tracked using Weights & Biases [1], GitHub [2] .

1 CNN

The ResNet-18 architecture from `torchvision` is used with `pretrained=False` and adapted for the CIFAR-10 dataset. The original 7×7 convolution with stride 2 is replaced by a 3×3 convolution with stride 1 to preserve spatial resolution for 32×32 images. The initial max-pooling layer is removed to avoid excessive downsampling. Finally, the fully connected layer is modified to output 10 classes, corresponding to the CIFAR-10 categories.

2 custom dataloader

This custom PyTorch dataset wraps the CIFAR-10 dataset and applies standard preprocessing. It converts images to tensors and normalizes them using CIFAR-10 mean and standard deviation. Separate instances are created for training and testing while sharing the same transform.
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3 FLOPs

Metric	Value
Parameters	~ 11.17 M
FLOPs (per forward pass)	~ 557.78 MMac

Table 1: CIFAR-10 ResNet-18

4 Training Setup

The model is trained for 25 epochs using the Adam optimizer with a learning rate of 1×10^{-3} . The cross-entropy loss function is used. Training is performed on GPU when available.

5 Gradient Flow Analysis

The gradient flow remains stable throughout training, with no signs of exploding or vanishing gradients. Consistent gradient magnitudes across layers indicate healthy backpropagation and effective parameter updates, contributing to smooth convergence.

6 Weight Update Flow Analysis

Weight update plots show controlled and consistent updates across convolution and batch normalization layers. The absence of abrupt spikes confirms stable learning dynamics, proper learning rate selection, and well-regularized training behavior.

References

7 Result

final training accuracy reported **99.508 percent**

- [1] Weights & Biases Experiment on CIFAR-10 Gradient and Weight Flow, <https://wandb.ai/b23cm1038-prom-iit-rajasthan/cifar10-gradient-weight-flow/workspace?nw=nwuserb23cm1038>
- [2] S. Changulani, “MLOps-B23CM1038 – Worksheet-1,” GitHub repository, <https://github.com/Sirin-890/MLOps-B23CM1038/tree/Worksheet-1>